Radio-frequency charge detection on graphene electron-hole double quantum dots – Supplementary Information –

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A. Device design

The device consists of a van der Waals heterostructure of BLG encapsulated between two crystals of hexagonal boron nitride (hBN), each approximately 50 nm thick, placed on a graphitic back gate (BG). Ohmic Cr/Au contacts are etched through the top hBN flake to the BLG. Fig. 1a of the Manuscript shows a false color scanning electron microscope (SEM) image of the metallic gate structure fabricated on top of the heterostructure. Split gates (SGs) and a graphite BG are used to open a band gap in the BLG, and to form conducting channels. In the wide channel (highlighted in blue), a QPC beneath the finger gate G0 (orange) acts as a capacitively coupled charge detector. This channel forms a T-junction with the narrow channel beneath the other finger gates (FGs), where the QDs will be localized. A layer of 15 nm Al_2O_3 is placed on top of the SGs, separating them from the Cr/Au FGs (orange outlines), which define and control the QDs and the QPC. These FGs have a center-to-center pitch of 140 nm and a width of 70 nm. The center of the wide channel has a lateral distance of 230 nm to the center of FG G2. To control the tunnel barriers between the QDs, the QPC charge detector and the reservoirs, an additional layer of FGs (green outlines) is placed on top, separated by another 15 nm Al_2O_3 [1]. For more details on the fabrication of the device, we refer to Ref. [2].

B. Schrödinger-Poisson model

The local charge carrier density $\rho[\phi(\mathbf{r}'), \partial_z\phi(\mathbf{r}')](\mathbf{r})$ can be simulated by solving Poisson's equation in dielectric media

$$\nabla(\varepsilon(\mathbf{r})\nabla\phi(\mathbf{r})) = \rho[\phi(\mathbf{r}'), \partial_z\phi(\mathbf{r}')](\mathbf{r}), \tag{1}$$

where $\varepsilon(\mathbf{r})$ represents the local dielectric constants of the materials, and $\phi(\mathbf{r})$ the electrostatic potential. If all permittivities were the same, i.e. $\epsilon(\mathbf{r}) = const.$, the gradient applied to it would be zero and we would get a single Laplacian on the left-hand side. However, the device consists of materials with different permittivities, hence $\epsilon(\mathbf{r})$ changes at interfaces. We account for this by the more general expression above, which can also accommodate tensor-valued ϵ necessary for different in-plane and out-of-plane permittivities. The voltages at the different gates serve as Dirichlet boundary conditions for $\phi(\mathbf{r})$; at all other boundary points we assume Neumann boundary conditions, $\frac{\partial\phi}{\partial\vec{n}} = 0$, where $\vec{n}$ is the normal vector to the surface of the structure at a given boundary point $\vec{r}_b$. We assume only BLG carries a charge carrier density, which depends on the local displacement field and its gradient. Consequently, $\rho[\phi(\mathbf{r}'), \partial_z\phi(\mathbf{r}')](\mathbf{r})$ depends on $\phi(\mathbf{r}')$ for the local shift in potential, and $\partial_z\phi(\mathbf{r}')$ for the local band gap. We use a local density approximation,

$$\rho[\phi, \partial_z\phi](\mathbf{r}) \approx \rho\left(\phi(\mathbf{r}), \partial_z\phi(\mathbf{r})\right) \approx -\int_0^{\phi(\mathbf{r})} D(\varepsilon'; \partial_z\phi(\mathbf{r}))\mathrm{d}\varepsilon' \tag{2}$$

with $D(\varepsilon'; \partial_z\phi(\mathbf{r}))$ being the density of states per unit area of BLG divided by its thickness (to obtain a volume density) at a given energy and displacement field $\partial_z\phi(\mathbf{r})$. To simplify further, we assume that the charge carrier density $\rho(x, y, z) = \rho(x, y)$ may only vary in-plane, but not out-of-plane within the 3.35 Å thickness of BLG. This allows replacing the upper integration bound in Eq. (2) $\phi(\mathbf{r})$, which varies as a function of three coordinates, by the mean evaluated at the top and bottom height of the BLG sheet at given in-plane coordinates,

$$\phi(\mathbf{r}) \to \overline{\phi}(x, y) = 0.5 \cdot (\phi(x, y, z_{\text{bot.}}) + \phi(x, y, z_{\text{top}}))$$

With a self-consistency loop, the charge carrier density in BLG can be numerically calculated for a given set of gate voltages. We start with an initial guess by solving the homogeneous problem, i.e. we set the right-hand side of Eq. (1) to zero and obtain ϕ_0. This first result ϕ_0 strongly overestimates the induced charge, since no screening is considered. To better condition the start of the self-consistency cycle, we therefore multiply ϕ_0 by a small number $\gamma \approx 0.025 \ll 1$ while leaving the gradient unchanged. Thus, we evaluate the first induced ρ_1 by

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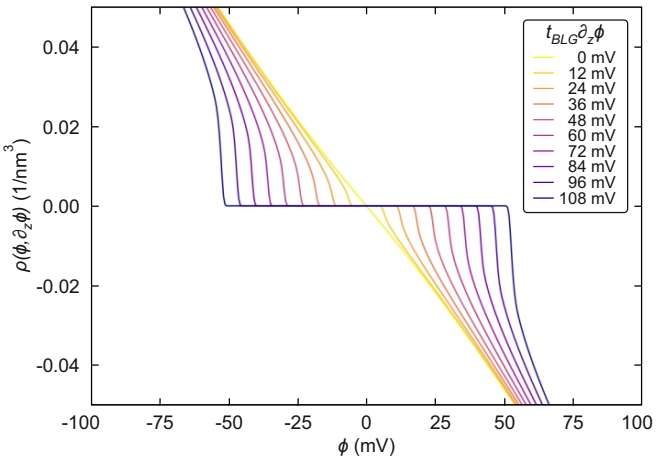


FIG. S1. Charge carrier density (Eq. (2)) as a function of potential ϕ for different gradients $\partial_z\phi$. The plateau at $\rho = 0$ emerges due to the opening band gap. Note the resulting strong nonlinearity at the band edge. We assume electron-hole symmetry.

inserting into Eq. (2)

$$\rho_1 = \rho[\gamma \cdot \phi_0, 1 \cdot \partial_z\phi_0]$$

This rescaling of ϕ_0 but not $\partial_z\phi_0$ is only done for the very first iteration to obtain an initial guess for ρ_1. For all subsequent iterations we try to find the root of the function

$$\rho_n - \rho[\phi_n, \partial_z\phi_n] = 0 \tag{3}$$

where ϕ_n is the solution of Eq. (1) with the right-hand side ρ_n, which we calculate with NGSolve [3]. At each iteration, the root finder provides a ρ_n for which we solve Poisson's Eq. (1), yielding ϕ_n. The latter we insert into Eq. (2) for the charge carrier density, and compare $\rho[\phi_n, \partial_z\phi_n]$ to ρ_n. We iterate this procedure until both values (approximately) agree and thus self-consistency is achieved. The difference between the inhomogeneity and the model in Eq. (3) is computed on a rectangular grid of $N = 200 \cdot 200 = 40 \cdot 10^3$ points inside the BLG crystal. Solving the non-linear Poisson's equation 1 amounts to finding a root of a vector-valued function with dimension of $N = 40 \cdot 10^3$.

The main challenge of the self-consistency Eq. (2) is the band gap in the density of states, the integrand of Eq. (2), which changes locally as a function of the out-of-plane derivative $\partial_z\phi(\mathbf{r})$. As depicted in Fig. S1, charge carriers are absent in the band gap. Two kinks show up in the charge carrier density at the Van Hove singularities due to the band edges. As a consequence, the right-hand side of Eq. (2) cannot be linearized locally at every point, thus the tested gradient methods, such as the "good" and "bad" Broyden methods to obtain the root of function 3, fail. However, we find that the derivative free spectral residual method [4] does work in finding the root of Eq. (3). We use the *scipy.optimize.root* implementation [5].

The average potential difference $\delta\phi$ at the QPC caused by a change in occupation depicted in the simulation curve in the main text (Fig. 3(f)) can be made more precise:

$$\delta\phi = \langle\phi\{V_{(i)}\}\rangle_{\mathrm{QPC}} - \langle\phi\{V_{(i)} + \delta V_{(i)}\}\rangle_{\mathrm{QPC}}$$

For this expression, we first calculate the electric potential ϕ at a specific configuration (for instance in Fig. 3(e) of the main text we go along a section of the $(1e, 0h)$ to $(2e, 0h)$ charge transition line in the charge stability diagram) but for different voltages $\{V_{(i)}\}$. Once this line $\{V_{(i)}\}$ is found, we perform another simulation for a small deviation $\delta V_{(i)}$ along this section in the charge-stability diagram. Experimentally, the QPC (and in consequence the measured step height) is sensitive to this change of the electric potential

$$\phi_{\mathrm{diff}} = \phi\{V_{(i)}\} - \phi\{V_{(i)} + \delta V_{(i)}\}$$

We map this function to a single scalar value by calculating the averages of the individual potentials inside BLG under the QPC ($\langle ... \rangle_{\mathrm{QPC}}$). This measure allows us to make general qualitative statements about the change of the magnitude of ϕ_{diff}. However, for a quantitative prediction one also needs to model the bound state spectrum of the BLG Hamiltonian and additionally account for screening due to the integer occupation of electrons and holes. This improvement of the model is computationally very challenging and left for future publications. Nevertheless, since the QPC is far away from the charges, we expect the qualitative trends of Fig. 3 (main text) to hold even in this case. We only expect corrections in the amplitude of ϕ_{diff} and in the immediate vicinity of the electron and hole bound states, leaving the functional form of the long-range electrostatic field intact.

material	ϵ
vacuum	1
polycarbonate	3.1
Al_2O_3	9.1
$hBN_{\parallel}$	4.98
$hBN_{\perp}$	3.03
$BLG_{\parallel}$	1.8
$BLG_{\perp}$	3

TABLE I. Scalar and tensor permittivities ϵ of the materials relevant in the experiment. For hBN and BLG we distinguish between in-plane $\epsilon_{\parallel}$ and out-of-plane $\epsilon_{\perp}$ permittivities (see [6] and [7]).

Fig. S2 shows a simulation of the potential ϕ_0 solving the homogeneous problem for the investigated device geometry and an exemplary set of gate voltages. Poisson's equation requires permittivities for each material. We use the scalar relative permittivities ϵ as given in Tab. I.

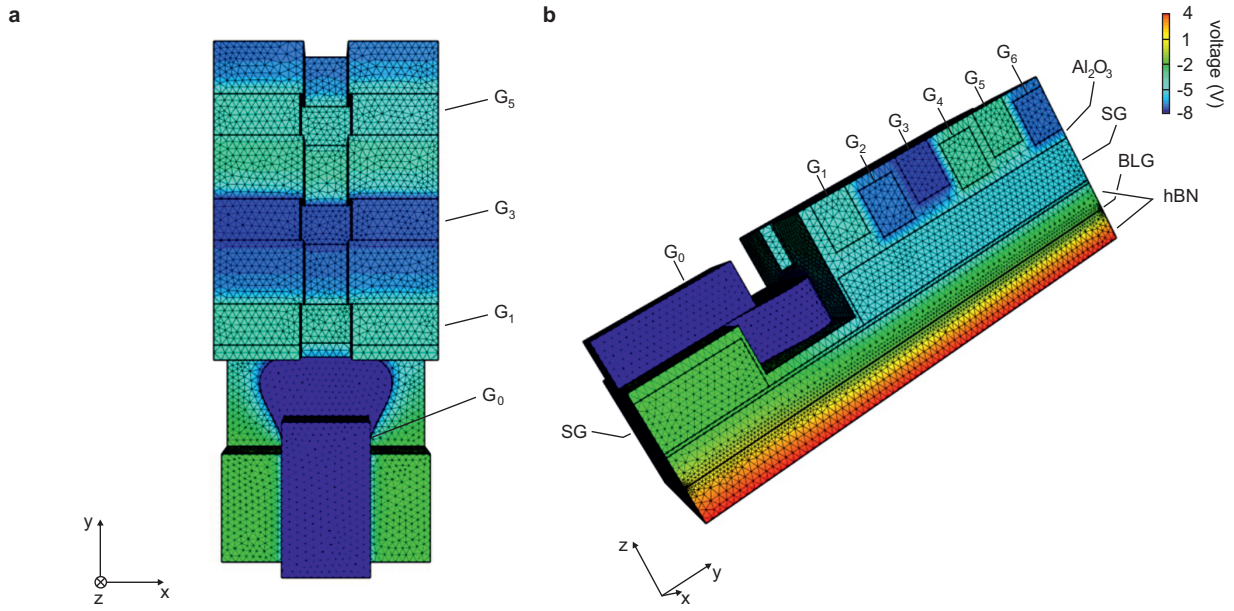


FIG. S2. (a) Top view and (b) side view of a 3D model of the used device, showing the heterostructure and the gate stack. The color code shows the voltages applied to the gates and the calculated potential throughout the device. A layer of vacuum covering the top gates filling out the simulation box is not shown for visibility.

C. Impact of the insulator thickness: a toy model

The distance between the bilayer and the metal gates strongly impacts the electrostatics of the device. While a thin insulating layer allows for a larger gate lever arm, it also increases screening, leading to much quicker decay of charge-induced potential variations with distance, making charge detection more difficult. Conversely, a thick insulating layer necessitates larger applied voltages, and thus potentially short-circuiting.

To understand how the insulator thickness affects the change of electrostatic potential ϕ inside the BLG sheet we numerically investigate a toy model. We leave an analytical calculation to a further study. We focus here on elucidating the connection between the sample geometry and the resulting electrostatic properties. We choose a simple model of a cylindrical gate on top of a bilayer sheet (see Fig. S3) to disentangle the different contributions to the electrostatic potential. Consider a BLG sheet of 300 nm$\times$100 nm area sandwiched between two layers of hBN of equal thickness w. Below the hBN-bilayer-hBN stack, there is a single bottom gate at voltage V_B, and on top there are two gates: a planar gate with a hole at a voltage of $-V_B$, and a cylindrical gate within this hole at voltage V_C.

To understand the influence of the hBN width w we consider the change from one to two electrons inside the dot, i.e., $Q_1 = 1e = -1$ and $Q_2 = 2e = -2$ (in atomic units) and fix a displacement field $\delta\phi_{int.lay.} = 30$ meV. These constraints uniquely determine $V_B(w)$, $V_{C,1}(w)$ and $V_{C,2}(w)$: for a given hBN thickness w, we determine (i) the $V_B(w)$ needed to arrive at $\delta\phi_{int.lay.}$ within the bulk BLG, evaluated at the left edge, far away from the circular gate. and (ii) the voltages $V_{C,1}(w), V_{C,2}(w)$ yielding the charges Q_1 and Q_2 below the circular gate. To obtain $\delta\phi_{12}$ numerically, we solve for ϕ_1 and ϕ_2 individually and then take the difference $\delta\phi_{12} = \phi_2 - \phi_1$. The non-linear source term $\rho[\phi, \partial_z\phi]$ is given by equation 2, just like the main text.

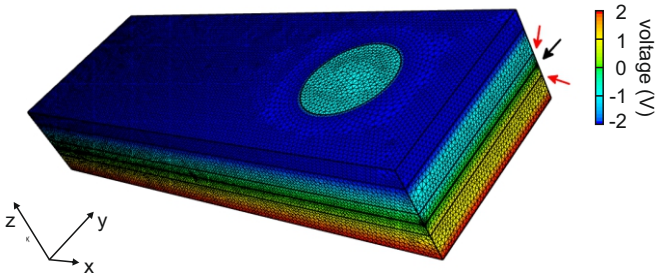


FIG. S3. Overview of the toy model. The electric potential ϕ which solves the homogeneous problem with charge density $\rho = 0$ is plotted for the entire structure, the color bar goes from $-2V$ (blue, top gate voltage $-V_B$) to $2V$ (red, bottom gate voltage V_B). The cylinder gate is held at $V_C = -1V$. Neumann boundary conditions apply $\frac{\partial\phi}{\partial\vec{n}} = 0$ to the remaining boundaries. The maximal triangle edge mesh size is chosen to be 2.5 nm. The hBN thickness depicted here is 20 nm, both layers of hBN are subdivided into two (the red arrow point to where this happens), such that the region close to the BLG sheet (which the black arrow points to) is resolved better.

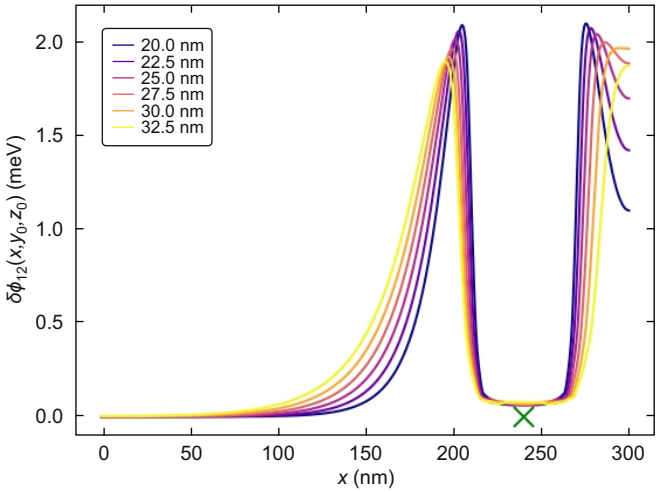


FIG. S4. $\delta\phi_{12}$ along the center line $y_0 = 50$ nm and $z_0 = w + 0.5 \cdot t_{BLG}$ for different hBN thicknesses w, where t_{BLG} is the BLG thickness. Exponential tails are elongated with increasing w. The center point of the cylindrical gate stands above the green cross.

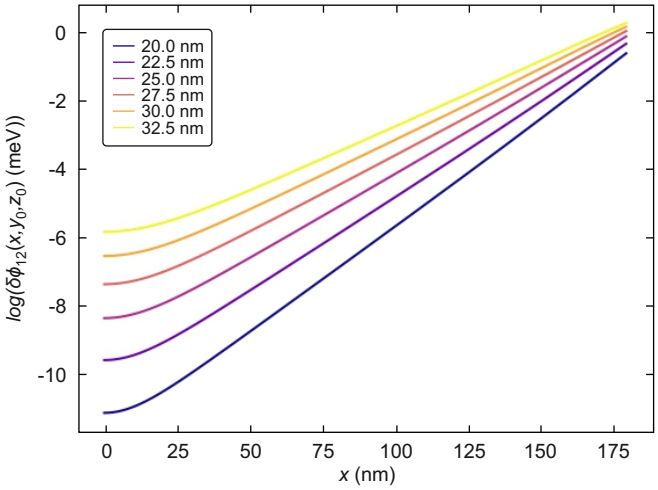


FIG. S5. Logarithmic plot of $\delta\phi_{12}$ after all exponential tails emerge at 175 nm. The slopes are the decay of each exponential. Toward $x = 0$ we see flattening of the logarithmic curves and thus flattening of $\delta\phi_{12}$, which is imposed by the Neumann boundary conditions.

$$\begin{aligned}
\delta\phi_{12}(\mathbf{r}) &= \phi_1(\mathbf{r}) - \phi_2(\mathbf{r}) \\
\nabla(\varepsilon(\mathbf{r})\nabla\phi_1(\mathbf{r})) &= \rho[\phi_1, \partial_z\phi_1] := \rho_1(\mathbf{r}) \\
\nabla(\varepsilon(\mathbf{r})\nabla\phi_2(\mathbf{r})) &= \rho[\phi_2, \partial_z\phi_2] := \rho_2(\mathbf{r}) \\
Q_1 &= \int d\vec{r}\rho_1(\mathbf{r}) = const. \\
Q_2 &= \int d\vec{r}\rho_2(\mathbf{r}) = const.
\end{aligned}$$

We consider hBN thicknesses $w = 20 \ldots 32.5$ nm in steps of 2.5 nm. We find that the potential difference $\delta\phi_{12}$ exponentially decreases away from the source (see Figs. S4 and S5), as a function of in-plane distance r in the BLG sheet, where the decay is decreasing with larger values of w. Intuitively, this occurs because the planar gates move away with increasing w, reducing their importance compared to the charge transition occuring in the BLG sheet.

Our numerical results suggest that, while the equations governing the quantum point contact (QPC) might be complicated, they result in exponential decay of the response, with the decay constant clearly decreasing with increasing w. Thus, increasing the insulator height w improves the signal, preventing the field lines emanating from the localized charge from being attenuated. These considerations should intuitively be true very far away from the source, where we approximately have a plate capacitor. Our numerical results remarkably predict that this exponential behavior also holds close to the gates.

While we performed these calculations for this simple toy model with a single source $\rho_{1,2}$, our reasoning is quite general and even holds if there is some common charge distribution unchanged by the variation of $V_{C,1} \to V_{C,2}$, such as a ρ_{left} (potentially acting as a charge detector) on the left side of the structure, as long as this is included in both charge distributions ρ_1 and ρ_2. From an experimental point of view, a larger w requires larger gate voltages, ultimately limiting the thickness w from above. We are currently working on a more detailed discussion and comparison with an analytical model.

One of the advantages of the T-shaped model of the device shown in the main text is that it is not an approximate plate capacitor, which is roughly the case in the usual double-channel geometry in other experiments. Thus, for the T-shaped geometry, there is no large top side gate between the QD and the charge detector, which according to our intuition obtained from the toy model of this section, should improve the detector signal $\delta\phi_{1,2}$.

While the model presented here solves the Poisson equation self-consistently, the quantum-mechanical part is based on the bulk density of states instead of a full solution of the Schrödinger (or Dirac) equation. A comprehensive treatment of this problem would need to simultaneously solve Poisson's equation and the quantum mechanical wave equation for the eigenstates of the electrons in the device to account for bound states, which poses a significant computational challenge. We conjecture here that the evident exponential decay of $\delta\phi_{12}$ with distance suggests that even for such an improved treatment of the quantum mechanical part, the plate capacitor solution of the left-side will determine the long-range electrostatic field. Thus, while quantitative predictions for, e.g., the height of the exponential tail and its decay constant may depend on the model employed, the qualitative long-range trend should remain unaffected. Ultimately, it is this long-range behavior which determines the sensitivity of the charge detector.

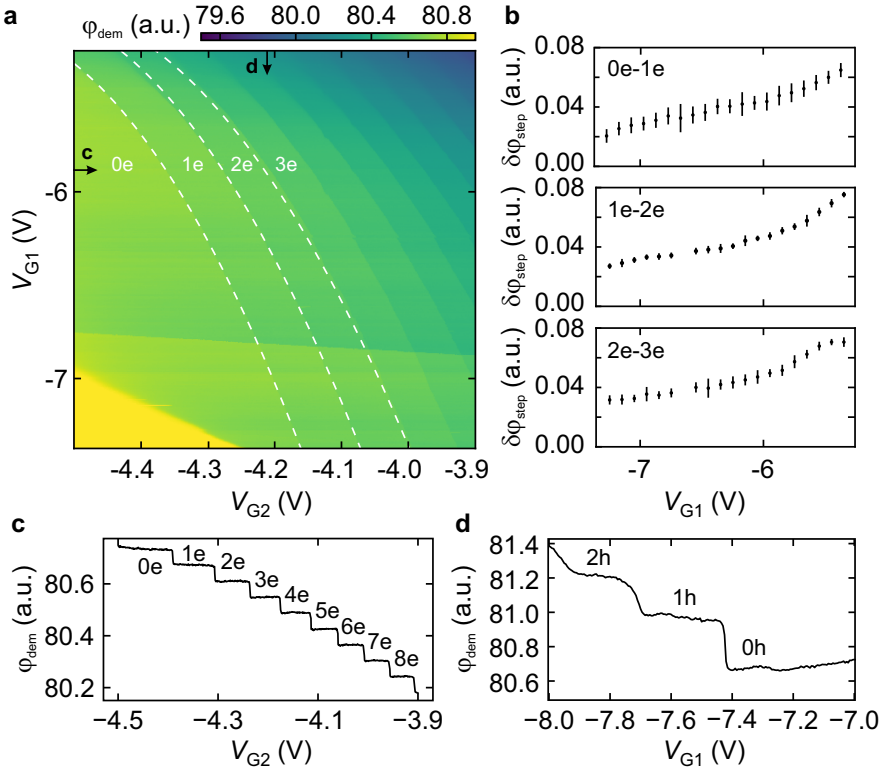


FIG. S6. (a) Charge stability diagram showing the phase φ_{dem} of the reflected RF signal as a function of the gate voltages V_{G1} and V_{G2}. (b) Step heights of the charge transitions 0e-1e, 1e-2e, and 2e-3e extracted from panel (a) (see white dashed lines). (c,d) Line cuts along the V_{G2}-axis and V_{G1}-axis (see arrows in panel (a)) showing the sequential filling of the electron QD.

D. Complementary data

Fig. S6a shows an extended range of the charge stability diagram in Fig. 3a of the manuscript. Multiple electron transitions are visible (see labels). The extracted step heights of the transitions 0e-1e, 1e-2e, and 2e-3e (dashed lines) are depicted in Fig. S6b. As can be seen, the qualitative behavior is equal for all three electron resonances. For the analysis shown in Fig. 3b and c of the manuscript, we picked the charge transitions 0h-1h and 1e-2e to present results with the best signal-to-noise ratio. Figs. S6c and d show cuts through the charge stability diagram along constant V_{G1} and V_{G2} showing sharp steps indicating the sequential filling of the electron and hole QD.

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